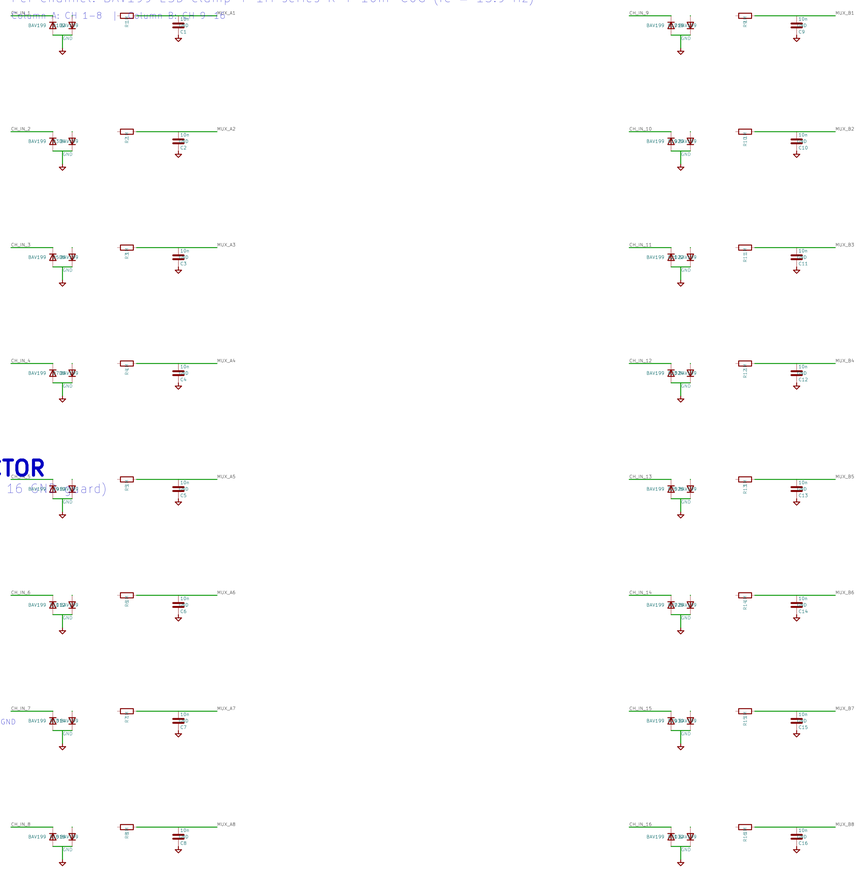
CircuitForge — 16-Channel Multiplexed Measurement System

ADuCM362 + ADA4530-1 TIA | 4 Relay-Selectable Ranges | 24-bit ADC

INPUT FILTERS (16 channels)

Per channel: BAV199 ESD clamp + 1M series R + 10nF C0G (fc = 15.9 Hz)

Each channel: 1 CH 1-8 | 1 channel 9-16

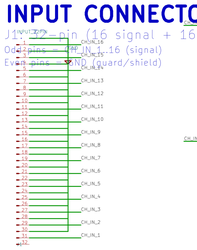


INPUT CONNECTOR

J1: 32-pin (16 signal + 16 guard)

Odd pins: CH_IN_1..16 (signal)

Even pins: GND (guard/shield)



Triax: center=signal, guard=driven, shield=GND

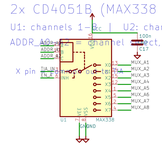
ANALOG MULTIPLEXER

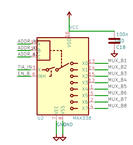
2x CD4051B (MAX338 equivalent) 8:1 mux

U1: channels 1-8 | U2: channels 9-16

ADDR_A/B/C: Channel select, EN_A/EN_B = mux enable

X pin 24 (VDD) to VCC





TRANSIMPEDANCE AMPLIFIER

U3: ADA4530-1 (20fA bias current)

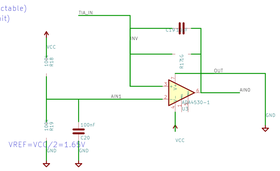
Rf/Cf: relay-selectable (10M / 100M / 1G+10pF / 10G+1pF)

VREF: 100k/100k divider = 1.65V mid-supply

Output: AIN0 (TIA) + AIN1 (VREF monitor)

Rf=1G default (relay-selectable)

Cf=10pF (stability, BW limit)



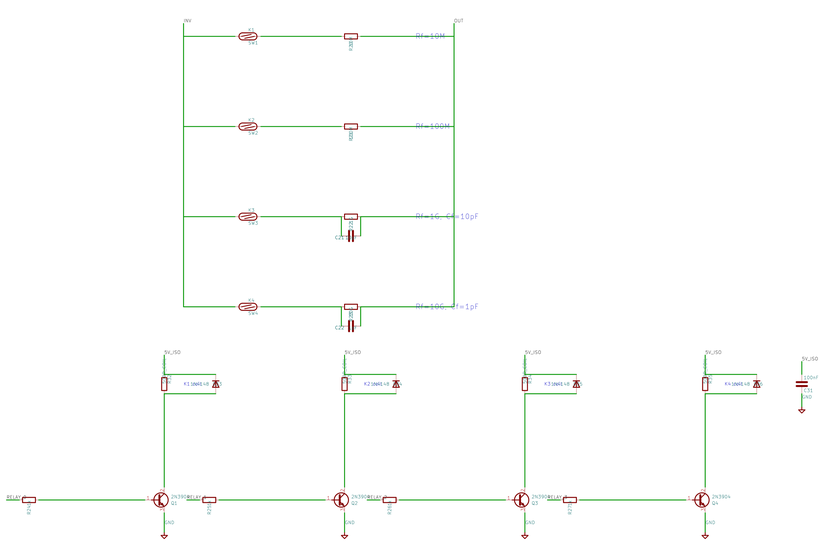
VREF=VCC/2=1.65V

RELAY RANGE LADDER

4x 109P reed relay + 2N3904 NPN drivers

K1: 10M (120nA FS) | K2: 100M (12nA FS) | K3: 1G+10pF (1.2nA FS) | K4: 10G+1pF (120pA FS)

RELAY_0-3 GPIO from MCU | 5V_ISO coil supply | 1N4148 flyback protection



Rb = 1k (Ic/Ib=5 at 100 mA, 6V gives 3.3mA base drive)

D33-D36: 1N4148 flyback diodes (relay coil back-EMF protection)

R32-R35: 500R relay coil resistance (109P-1-A-5/1 reed relay)